

Deterministic Short-Term Coordination of Water Distribution and Distributed Energy Assets Using Mixed-Integer Linear Programming

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Abstract

Water storage makes pumping an electrically flexible demand, but the availability of this flexibility depends on hydraulic service requirements and electrical network limits. This paper presents a deterministic mixed-integer linear programming implementation for hourly coordination of a water distribution network, photovoltaic generation, battery storage, and grid exchange. The formulation combines junction mass conservation, tank inventory dynamics, piecewise-linear pipe and pump characteristics, nodal active and reactive power balances, and linearized voltage and current constraints. Pump status links the two networks through an electrical demand calibrated from an initial hydraulic simulation. Every operating interval updates both storage inventories, and terminal replenishment prevents apparent savings obtained by depleting water or battery reserves. A reproducible illustrative study uses the supplied small water network and electrical feeder, with controlled comparisons of tariffs and pumping decisions. In the illustrative comparison, coordination lowers modeled daily cost by 0.07% relative to the recorded independently selected pumping schedule, while leaving total modeled import unchanged. Nonlinear replay preserves sampled pressure and voltage limits but identifies a terminal water-reserve shortfall, demonstrating the need to check optimized decisions beyond MILP feasibility. The implementation provides a transparent baseline for evaluating water–energy coordination and for subsequent improvements to pump–power modeling, simulation-based validation, and uncertainty treatment.

Keywords: water–energy nexus; pump scheduling; battery storage; mixed-integer linear programming; distribution networks; reproducibility.

1 Introduction

A water distribution operator can shift some electricity consumption by filling storage tanks before expensive periods and drawing down those inventories later. This possibility creates a scheduling problem: electricity procurement can be reduced or moved only if water demands, pressure requirements, and storage limits remain satisfied. When pumping shares an electrical network with other loads, photovoltaic (PV) generation, and batteries, its preferred schedule also depends on feeder capacity and the timing of other resources. Optimizing each asset independently can miss feasible exchanges of flexibility between them.

The underlying coordination problem is not new. Oikonomou and Parvania [1] explicitly study the flexibility of water distribution operations in day-ahead power-system scheduling. At the component level, the hydraulic constraints required to describe pipes, pumps, and tanks introduce nonlinearities and discrete operating states. Thomas and Sela [2] demonstrate how such features can be represented within an accessible mixed-integer linear programming (MILP) framework. On the electrical side, Morvaj et al. [3] integrate approximations of distribution-grid constraints into distributed-energy design and operation. These works motivate a common scheduling model in

which water-service constraints determine feasible pumping decisions and the electrical model prices their consequences.

This paper develops and documents an executable deterministic baseline for that integration. The intended decision maker is a single coordinating operator with access to the network descriptions, expected demands, electricity tariffs, and asset limits. The objective is daily electricity procurement cost, after export revenues, with water and electricity service treated as constraints. The contribution is an explicit formulation and a reproducible implementation that connects hydraulic scheduling to bus-level electrical demand, accounts consistently for both water and battery inventories through the end of the horizon, and separates solver feasibility from nonlinear simulation evidence. It is an implementation study rather than a claim of a new general-purpose hydraulic approximation or a new decomposition algorithm.

The scope is deliberately limited to fixed-speed water-distribution pumps, cylindrical tanks, fixed network topology, PV, and electrical storage. All input profiles are known for the optimization horizon. There are no scenarios, recourse decisions, wind generators, hydropower turbines, heat balances, electricity-market bidding decisions, or decentralized price-coordination algorithms in the reported model. The numerical study is an illustrative computational experiment with supplied network files, rather than an empirical study of a particular community.

2 Related work and modeling position

Water-network scheduling requires both conservation laws and device operating logic. MILPNet [2] provides a formulation for hydraulic elements and controls and compares approximations with EPANET simulations. The present implementation adopts the general use of piecewise-linear hydraulic relations and binary pump decisions, while using a smaller supported component set and explicit inventory constraints. It does not reproduce every MILPNet control or inherit the accuracy of MILPNet’s published benchmarks merely by using similar approximations.

Oikonomou and Parvania [1] connect hydraulic operating constraints to electricity-system flexibility and day-ahead operation. The present work instead studies a centralized local scheduling problem with explicit PV and battery decisions on a small feeder. That choice avoids the additional market and operator-interaction assumptions needed to interpret a flexibility bid. Morvaj et al. [3] show how distributed-energy scheduling can incorporate linearized AC power flows and branch-current constraints. Here, those modeling ideas are specialized to fixed assets and a balanced, fixed-topology electrical network. Asset sizing, building thermal dynamics, and multiple energy carriers are outside the formulation.

EPANET supplies a nonlinear hydraulic reference model [4]; WNTR provides its Python network interface and simulation tools [5]. OpenDSS provides the electrical network description and a separate power-flow simulation capability [6]. These simulators have a different role from the MILP solver. An initial simulation can calibrate an approximation, while a replay of optimized controls can assess it. Agreement obtained after fixing variables or linearizing at a simulator’s known operating point is a diagnostic of implementation consistency, not an independent test of scheduling accuracy.

3 Problem definition and notation

The horizon consists of intervals $\mathcal{T} = \{0, \dots, T - 1\}$ and inventory states $\{0, \dots, T\}$. The implementation uses one-hour intervals, so $\Delta t_h = 1$ h and $\Delta t_s = 3600$ s. Water link flows are in m^3/s , hydraulic heads in m, active power in kW, reactive power in kvar, and battery energy in kWh. Electrical voltages and currents are per-unit, and angles are radians. The three-phase electrical

power base S_B is expressed numerically in kVA, giving active and reactive power scales of kW and kvar.

A directed water graph contains junctions $\mathcal{J}$, tanks $\mathcal{K}$, reservoirs, and links $\mathcal{L}$, including pumps $\mathcal{P}$. Link orientation defines the sign of flow; ordinary pipes permit both signs. Let A_{nl}^w equal +1 for a link entering water node n , -1 for a link leaving it, and zero otherwise. The analogous incidence matrix for electrical buses $\mathcal{B}$ and lines $\mathcal{E}$ is A^e . Grid exchange is available only at the designated connection buses $\mathcal{G} \subseteq \mathcal{B}$.

Table 1: Principal variables and parameters. Bounds and asset capacities are data, not design decisions.

Symbol	Meaning	Unit
q_{lt}, h_{nt}	Water-link flow and total hydraulic head	$\text{m}^3/\text{s}, \text{m}$
u_{pt}	Pump on/off decision	binary
d_{nt}^w, z_n	Junction demand and elevation	$\text{m}^3/\text{s}, \text{m}$
A_k, h_k^0	Tank cross-sectional area and initial head	m^2, m
p_{bt}^g, p_{bt}^x	Grid import and export	kW
$p_{bt}^{pv}, \bar{p}_{bt}^{pv}$	PV dispatch and available PV	kW
$p_{bt}^{ch}, p_{bt}^{dis}$	Battery charge and discharge at its AC bus	kW
e_{bt}	Battery energy at interval boundary t	kWh
p_{lt}, r_{lt}	Oriented line active and reactive power	kW, kvar
v_{bt}, θ_{bt}	Voltage magnitude and phase angle	pu, rad
i_{lt}^r, i_{lt}^i	Real and imaginary line-current components	pu
$\bar{p}_p$	Calibrated electrical draw of an operating pump	kW
c_t^{buy}, c_t^{sell}	Electricity purchase and sale prices	currency/kWh

4 Coordinated MILP formulation

4.1 Water conservation and inventory

For each junction and interval, inflow minus outflow must equal the prescribed demand:

$$\sum_{l \in \mathcal{L}} A_{nl}^w q_{lt} = d_{nt}^w, \quad n \in \mathcal{J}, t \in \mathcal{T}. \quad (1)$$

Demand is summed over all demand categories and evaluated using the network's time patterns. A minimum service pressure is imposed as

$$h_{nt} - z_n \geq \underline{\pi}_n, \quad n \in \mathcal{J}, t \in \mathcal{T}. \quad (2)$$

The demand-driven assumption means that these equations do not describe pressure-dependent consumption or leakage. Reservoir total heads follow their prescribed input time series.

For cylindrical tank k , the inventory is represented by its total head. Conservation over every interval gives

$$h_{k,t+1} = h_{kt} + \frac{\Delta t_s}{A_k} \sum_{l \in \mathcal{L}} A_{kl}^w q_{lt}, \quad k \in \mathcal{K}, t \in \mathcal{T}, \quad (3)$$

$$h_{k0} = h_k^0, \quad \underline{h}_k \leq h_{kt} \leq \bar{h}_k, \quad k \in \mathcal{K}, t = 0, \dots, T, \quad (4)$$

$$h_{kT} \geq h_k^0, \quad k \in \mathcal{K}. \quad (5)$$

The bounds include tank elevation, and the terminal condition preserves the initial water reserve. This condition can be changed for a different planning application, but all compared schedules must use the same terminal requirement. An hour- $T - 1$ withdrawal therefore changes h_{kT} and cannot disappear from the accounting. The model constrains interval-end volumes; it does not simulate every within-hour tank-control event.

4.2 Pipe and pump characteristics

For an ordinary pipe $l = (i, j)$, the Hazen–Williams head-loss relation is

$$h_{it} - h_{jt} = f_l(q_{lt}), \quad f_l(q) = K_l \operatorname{sgn}(q)|q|^{1.852}, \quad K_l = \frac{10.66L_l}{C_l^{1.852}D_l^{4.87}}, \quad (6)$$

where L_l and D_l are in metres and C_l is roughness. The optimizer replaces f_l with interpolation over bounded breakpoints. Bounds are based on initial simulated flow magnitudes or explicit overrides; thus an excessively narrow domain can exclude otherwise feasible schedules. The piecewise approximation is an equality surrogate for the nonlinear relation, not a global relaxation with a guaranteed hydraulic-error bound.

For a fixed-speed head pump with a one-point or three-point curve, EPANET supplies coefficients of the power-law characteristic $g_p(q) = a_p - b_p q^{\nu_p}$. A single design point implies a quadratic curve; two-point and other multipoint curves use linear interpolation and extrapolation of their supplied points. A piecewise representation over the nonnegative head-flow range gives a nonnegative head-gain variable γ_{pt} . Pump operation is enforced by

$$0 \leq q_{pt} \leq \bar{q}_p u_{pt}, \quad (7)$$

$$-M_p(1 - u_{pt}) \leq h_{jt} - h_{it} - \gamma_{pt} \leq M_p(1 - u_{pt}), \quad p = (i, j). \quad (8)$$

Thus a closed pump carries no flow and its two endpoint heads are decoupled; an operating pump follows its approximated head curve. Finite flow and head bounds are necessary both for physical interpretation and for valid disjunction constants M_p .

For clarity, all univariate piecewise relations use the same construction. Given points (x_s, y_s) , $s = 0, \dots, S$, introduce nonnegative weights λ_s and segment binaries δ_s , $s = 0, \dots, S - 1$:

$$x = \sum_{s=0}^S \lambda_s x_s, \quad y = \sum_{s=0}^S \lambda_s y_s, \quad \sum_{s=0}^S \lambda_s = 1, \quad (9)$$

$$\sum_{s=0}^{S-1} \delta_s = 1, \quad \lambda_s \leq \delta_{s-1} + \delta_s, \quad \delta_s \in \{0, 1\}. \quad (10)$$

Undefined endpoint binaries are zero. For a pump, the right-hand sides of both sum-to-one constraints are replaced by u_{pt} , so that both flow and the auxiliary head gain become zero when the pump is off. These constraints allow interpolation only between adjacent breakpoints and can be passed to an ordinary MILP solver without requiring native special-ordered-set support. The reported example contains no valves. General valve and automatic-control behavior must be validated separately before extending the case-study conclusions to networks containing those devices.

4.3 Electrical assets and storage

At bus b , PV can be curtailed within its known availability,

$$0 \leq p_{bt}^{pv} \leq \bar{p}_{bt}^{pv}. \quad (11)$$

Battery charging and discharging are defined at the AC bus. Their efficiencies appear only in the inventory equation:

$$e_{b,t+1} = e_{bt} + \Delta t_h \left(\eta_b^{ch} p_{bt}^{ch} - \frac{p_{bt}^{dis}}{\eta_b^{dis}} \right), \quad (12)$$

$$e_{b0} = e_b^0, \quad 0 \leq e_{bt} \leq \bar{e}_b, \quad t = 0, \dots, T, \quad (13)$$

$$e_{bT} \geq e_b^0. \quad (14)$$

Binary decisions y_{bt}^{ch} and y_{bt}^{dis} enforce

$$0 \leq p_{bt}^{ch} \leq \bar{p}_b^{ch} y_{bt}^{ch}, \quad 0 \leq p_{bt}^{dis} \leq \bar{p}_b^{dis} y_{bt}^{dis}, \quad y_{bt}^{ch} + y_{bt}^{dis} \leq 1. \quad (15)$$

A bus without storage has zero storage capacity and zero charge/discharge bounds. Grid import and export use analogous bounded binaries with mutually exclusive modes. Both exchanges are zero at buses outside $\mathcal{G}$. PV and battery exchanges are modeled at unity power factor, with no reactive dispatch. The model does not account for battery degradation.

4.4 Water–electricity coupling and bus balances

Let $\mathcal{P}_b$ contain the pumps electrically connected at bus b . Their modeled electrical demand is

$$p_{bt}^{pump} = \sum_{p \in \mathcal{P}_b} \bar{p}_p u_{pt}. \quad (16)$$

For a calibration simulation with positive pump head gain $\Delta h_{p\tau}^{sim}$, the coefficient is

$$\bar{p}_p = \frac{9.81}{\eta_p} \text{mean}_{\tau \in \mathcal{O}_p} \left(q_{p\tau}^{sim} \Delta h_{p\tau}^{sim} \right), \quad (17)$$

where $\mathcal{O}_p$ contains operating samples and η_p is the assumed wire-to-water efficiency. The factor 9.81 converts hydraulic power to kW for water density 1000 kg/m³. This is a constant on-state power approximation. In particular, changing optimized flow or head while leaving $u_{pt} = 1$ does not change the modeled electrical draw. Pumping cost therefore reflects on-hours and their timing, rather than the full flow–head-dependent energy requirement.

Let d_{bt}^e and d_{bt}^r denote non-pump active and reactive loads, and r_{bt}^g grid reactive injection. A single active balance at each bus includes every local asset and network exchange:

$$p_{bt}^g - p_{bt}^x + p_{bt}^{pv} + p_{bt}^{dis} - p_{bt}^{ch} + \sum_{l \in \mathcal{E}} A_{bl}^e p_{lt} = d_{bt}^e + p_{bt}^{pump}, \quad (18)$$

$$r_{bt}^g + \sum_{l \in \mathcal{E}} A_{bl}^e r_{lt} = d_{bt}^r. \quad (19)$$

Reactive grid injection is available only at grid buses and has no separate source-capability bound; feeder-current limits still restrict transmitted reactive power. Pumps are modeled at unity power factor in the electrical coupling. A non-unity pump power factor would require an additional reactive-demand term. The nodal balances permit local generation or storage to supply other buses through the feeder, with consistent flow orientation.

4.5 Linearized feeder constraints

Each line $l = (i, j)$ has series impedance $R_l + jX_l$ in per-unit and admittance $G_l + jB_l$, with

$$G_l = \frac{R_l}{R_l^2 + X_l^2}, \quad B_l = -\frac{X_l}{R_l^2 + X_l^2}. \quad (20)$$

A first-order expansion about $v = 1$ pu and equal angles gives

$$p_{lt} = S_B [G_l(v_{it} - v_{jt}) - B_l(\theta_{it} - \theta_{jt})], \quad (21)$$

$$r_{lt} = S_B [-B_l(v_{it} - v_{jt}) - G_l(\theta_{it} - \theta_{jt})]. \quad (22)$$

The reference bus has $v = 1$ and $\theta = 0$, and other voltage magnitudes obey their configured operating band. The first-order line model has no real-power losses: each line has one signed flow and its receiving-end flow is its negative. This approximation omits losses, shunt charging, phase imbalance, transformer tap changes, and higher-order AC terms. A mathematical optimum of this model is consequently not a certificate of AC feasibility.

The associated current components are

$$i_{lt}^{re} = G_l(v_{it} - v_{jt}) - B_l(\theta_{it} - \theta_{jt}), \quad (23)$$

$$i_{lt}^{im} = B_l(v_{it} - v_{jt}) + G_l(\theta_{it} - \theta_{jt}). \quad (24)$$

Introduce nonnegative $a_{lt} \geq \pm i_{lt}^{re}$ and $b_{lt} \geq \pm i_{lt}^{im}$. Both squared magnitudes are approximated by piecewise-linear secants of x^2 on $[0, \bar{i}_l]$:

$$s_{lt}^a = \text{PWL}(a_{lt}^2), \quad s_{lt}^b = \text{PWL}(b_{lt}^2), \quad s_{lt}^a + s_{lt}^b \leq \bar{i}_l^2. \quad (25)$$

The notation denotes interpolation of the square function, not a quadratic constraint. Since each secant lies above x^2 , this construction is conservative for the magnitude of the *linearized* current. On a segment of width Δx , the maximum square-function interpolation excess is $\Delta x^2/4$. Finer segments reduce this conservatism at the cost of more binary variables. They do not correct the error in the underlying linearized AC model.

4.6 Objective and feasibility interpretation

The coupled objective is

$$\min C = \Delta t_h \sum_{t \in \mathcal{T}} \sum_{b \in \mathcal{G}} \left(c_t^{buy} p_{bt}^g - c_t^{sell} p_{bt}^x \right). \quad (26)$$

Pumping enters this cost through the electrical balance and is charged once. There is no additional arbitrary water-flow cost in the coupled problem. Water-only and energy-only modes are useful diagnostics, but their objective values must not be interpreted as directly comparable community procurement costs.

The reported service-feasible formulation uses exact junction conservation. If diagnostic demand slacks are explicitly enabled, equation (1) becomes $\sum_l A_{nl}^w q_{lt} + s_{nt}^+ - s_{nt}^- = d_{nt}^w$ and a large penalty is added. Such a solution is useful for locating infeasibility, but any nonzero slack must be reported as a service-balance violation. A solver time limit likewise requires explicit distinction between an optimal solution, a feasible incumbent with a remaining gap, and a run with no usable solution.

5 Implementation and experimental protocol

The Python implementation uses Pyomo [7, 8] to assemble a shared model from optional water and energy components. EPANET-format files are read through WNTR. Electrical buses, loads, line parameters, and base quantities are extracted from the supplied OpenDSS circuit. Configuration files specify the horizon, assets, tariffs, pump-to-bus assignments, model bounds, and solver options. This separates physical network input from the economic experiment. The reported study uses Gurobi 12.0.3 through Pyomo 6.9.4, with Python 3.12.10 on macOS ARM64. The relative mixed-integer optimality-gap tolerance is 10^{-4} : an optimal termination denotes certification at that tolerance, rather than an assertion that every reported gap is exactly zero. Dependency versions and source/input hashes are recorded with the numerical outputs.

The workflow first validates configuration and prepares deterministic profiles, then runs a hydraulic calibration simulation, builds the MILP, solves it, and writes operational schedules and summary information. The source distinguishes calibration and diagnostic operating-point linearization from the piecewise scheduling formulation. Published scheduling results must be generated using the latter, unless a different approximation is explicitly identified. Solver status and the presence of an incumbent are checked before numerical solution extraction.

The experiment asks whether jointly choosing pump operation and electrical dispatch changes procurement cost when compared with a feasible fixed pumping schedule under the same asset availability and tariff. A tariff comparison also examines how schedules respond to time-varying prices. Only within-tariff comparisons with identical constraints estimate the scheduling benefit: the difference between bills under two different tariffs combines an operating response with a change in prices. Likewise, comparing a case with a battery to one without it measures both a change in the available assets and their scheduling.

All numerical comparisons should retain the same demand profiles, initial inventories, terminal requirements, PV availability, pump-power coefficients, and grid-connection assumptions. The baseline’s fixed decisions are imposed within the same feasible set, with its remaining electrical dispatch optimized. A flat-price water objective can admit multiple optimal pumping schedules, so the reported comparison is relative to the particular recorded baseline schedule. It does not estimate an average over possible uncoordinated operating policies. With exact solves, allowing the fixed decisions to vary cannot increase the optimal objective. Failure of this ordering would indicate unequal assumptions, solver gaps, or a formulation/reporting error, rather than negative value of flexibility.

Reported metrics include grid import energy, procurement cost, pump on-hours, battery charge and discharge energy, minimum junction pressure, tank inventory extrema and terminal values, voltage extrema, line loading, solver termination, and constraint residuals. Inventory residuals use all T intervals. For a reproducible hydraulic replay, the optimized pump schedule must be applied to the original nonlinear network with consistent demand timing and control handling; pressure, flows, and terminal inventory are then compared. A power-flow replay must include optimized PV, storage, and pump injections. An unmodified simulation of the input networks is only a reference trajectory, and is not described as validation of the optimized solution.

6 Numerical experiments

6.1 Illustrative case and comparison protocol

The supplied SNET water network contains two junctions, two pipes, one fixed-speed pump, one reservoir, and one cylindrical tank. The tank has a 12 m diameter, elevation 25 m, water-level bounds

0–6 m, and initial level 3 m. Nominal water demand is 25 L/s with the supplied 24-hour pattern. A 20 m junction-pressure floor and terminal tank replenishment are enforced. The head-pump design point is 50 L/s at 40 m. Pipe head losses use 12 segments over each signed flow domain; the pump uses six segments. The electrical example is the supplied two-bus, 115 kV circuit, with a 1000 kW nominal load and its supplied daily multipliers. This lightly loaded illustrative circuit is not a realistic low-voltage community feeder or a stress test of network congestion.

One 100 kW PV installation and one 200 kWh battery are located at the load bus. The battery has 50 kW charge/discharge limits, 0.95 efficiencies, and an initial and minimum terminal energy of 100 kWh. The synthetic PV availability peaks at hour 12. Synthetic purchase prices are 0.08 for hours 0–6 and 21–23, 0.16 for hours 7–15, and 0.32 for hours 16–20, in currency/kWh; exports earn zero. These profiles are controlled experimental inputs, not observations or an actual utility tariff. Pump efficiency is 0.75; the calibrated on-power is 25.948 kW. All intervals last one hour.

The independent baseline first minimizes water-side pumping energy under a constant price, then fixes that pump schedule while optimizing electrical asset dispatch under the time-varying prices. Joint scheduling releases the pump binaries and optimizes the same electrical cost with identical network, asset, and terminal constraints. The baseline uses the schedule returned by the water-only solve; alternative equally energy-efficient pumping schedules may have different time-varying electricity costs. The comparison therefore measures improvement over this explicit reproducible baseline, not a universal bound on the value of coordination. A separate flat-price joint case is a sensitivity run and is not directly comparable in monetary cost.

6.2 Optimization outcomes

Table 2 reports the saved solver outputs. Joint scheduling reduces daily electricity procurement from 2943.26 to 2941.18 relative to the independent baseline, a difference of 2.08 (0.07%). Values refer to the MILP approximation. The pump schedule and tank trajectory are shown in Figure 1. The initial water and battery reserves are preserved in the optimization.

Table 2: Computed schedules. Costs use synthetic currency units; flat-price cost uses a different tariff. Time is elapsed solver-call time on the recorded machine.

Case	Cost	Pump on (h)	Import (kWh)	Solve (s)
Independent	2943.26	9	16954.05	0.07
Joint	2941.18	9	16954.05	4.25
Joint, flat price	1693.35	9	16933.53	2.66

The compared joint model has 3219 variables and 2598 constraints. All three coupled solves and the independent water solve report optimal termination. The independent water-scheduling solve took 0.43 s. The run directory records full-precision dispatch, inventory trajectories, solver logs, algebraic residuals, input configuration, dependency versions, and input/source hashes. Timing includes solver-interface overhead and is not an isolated solver benchmark.

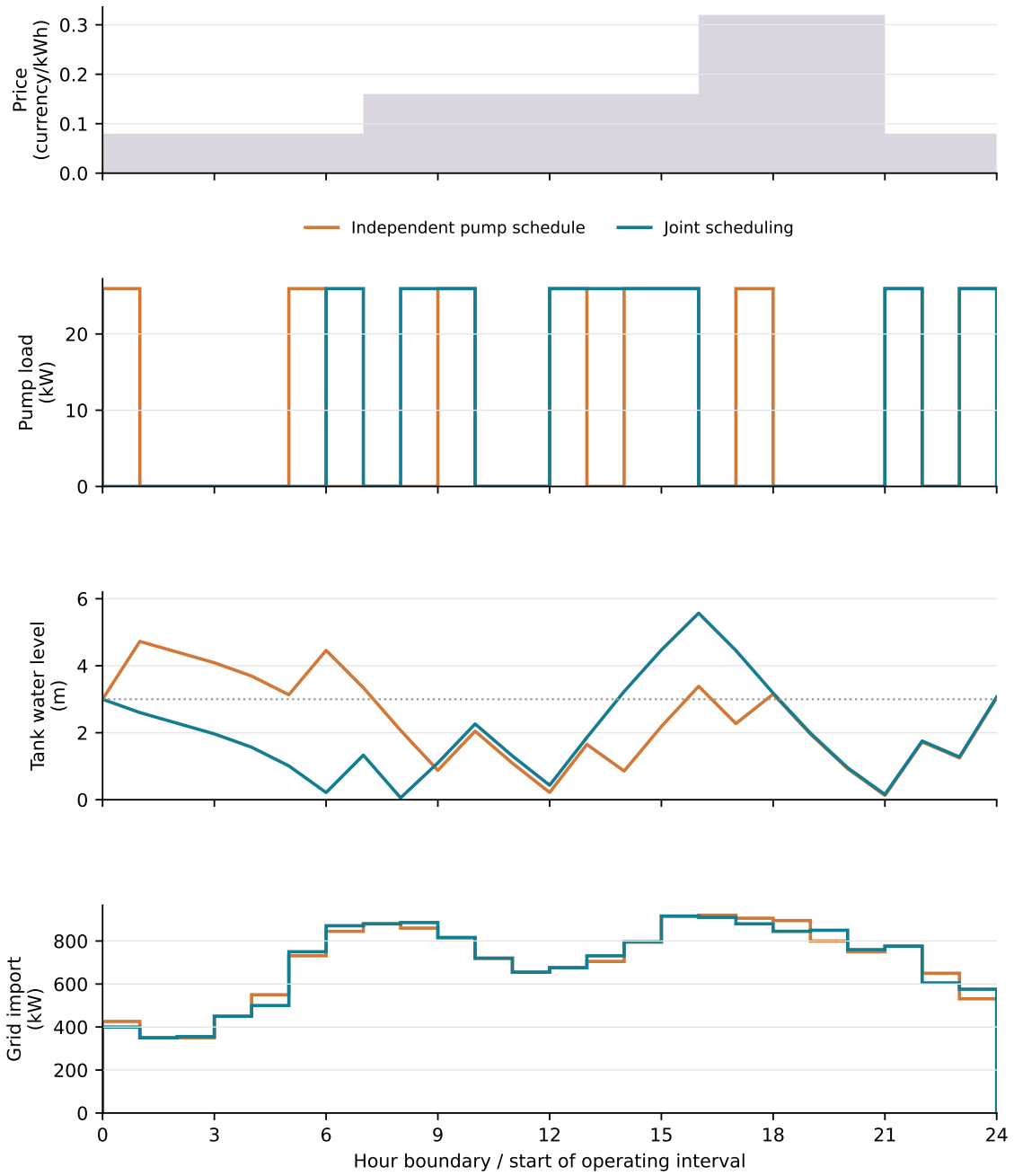


Figure 1: Synthetic purchase prices and optimized schedules for the independent baseline and joint coordination. Tank points represent inventory boundaries; power values apply over the following hourly interval.

6.3 Independent nonlinear replay

Each optimized pump schedule is replayed in EPANET using a five-minute hydraulic time step and hourly reports, replacing the original level controls with the hourly commands. Electrical net bus injections are replayed as balanced constant-power snapshots in OpenDSS. The replay retains the original circuit’s nonlinear AC behavior, whereas the MILP neglects series losses and shunt effects. The electrical replay retains the same fixed on-state pump-power approximation; it does not validate pump-energy calibration. These comparisons assess the proposed operating schedules rather than fitting an optimization model to an already known replay solution.

For the independent schedule, the minimum replay junction pressure is 25.06 m, with 0 pressure violations and 0 pump-status mismatches at report times. Maximum tank-head and flow discrepancies are 0.154 m and 0.466 L/s, respectively. The replay terminal tank head changes by -0.114 m relative to the initial state. All electrical snapshots converge; the maximum voltage discrepancy is 0.000338 pu and peak line loading is 1.31%.

For the joint schedule, the minimum replay junction pressure is 25.07 m, with 0 pressure violations and 0 pump-status mismatches at report times. Maximum tank-head and flow discrepancies are 0.135 m and 0.437 L/s, respectively. The replay terminal tank head changes by -0.067 m relative to the initial state. All electrical snapshots converge; the maximum voltage discrepancy is 0.000337 pu and peak line loading is 1.29%.

The replay loses some terminal water reserve despite MILP replenishment. The hourly hydraulic approximation therefore does not preserve the terminal requirement exactly; a reserve margin or a simulation-informed refinement is needed before operational use.

The hourly hydraulic control discretization and fixed pump-power coefficient remain approximation sources even when more pipe breakpoints are used. Hourly report checks do not certify feasibility between reports. These results support use as a scheduling prototype with explicit simulation checks; they do not constitute certification for field operation.

7 Discussion and limitations

The computed cost difference arises from timing rather than reduced modeled electricity use: the independent and joint schedules have the same total grid import and pump on-hours. Its small size should be interpreted in the context of the modest pump load relative to the much larger non-pump electrical demand. The selected feeder is lightly loaded, so these experiments do not demonstrate a benefit from relieving network congestion.

The formulation gives tank and battery reserves the same accounting role: each can move consumption across time, but neither can supply net energy or water by an unrecorded final-period withdrawal. Terminal replenishment is therefore part of the economic comparison, not a cosmetic postprocessing condition. Storage flexibility is limited by service pressure, flow characteristics, charging rates, efficiency, and the finite electrical connection.

The fixed on-state pump coefficient is the principal coupling approximation. It keeps the water–electricity interface linear and makes the effect of pump timing directly visible. It can also favor a schedule whose simulated flow and head depart from the calibration trajectory. Consequently, modeled pump-energy savings should be interpreted as savings under that coefficient until hydraulic replay flows and heads are converted to electrical consumption. The electrical replay in this study retains the same calibrated pump-power coefficient, so it assesses the feeder approximation without validating the pump-energy model. A next implementation step is a validated piecewise relation for electrical pump power as a function of hydraulic operating conditions; variable-speed operation would need additional speed-dependent characteristics.

Several further limitations constrain the numerical evidence. A small synthetic water network and a short feeder do not establish scalability or performance under congestion, diverse topology, or real operator data. Coarse hydraulic interpolation can bias pressures and volumes, while flow domains derived from a reference schedule can exclude feasible operations outside that domain. The linearized electrical model is most defensible near its flat-voltage expansion point; conservatism in equation (25) does not compensate for nonlinear AC error. Hourly decisions can miss subhourly tank-control events, motor starts, and short voltage excursions. Water quality, leakage, battery wear, pump switching costs, equipment outages, and forecast uncertainty are absent.

Finally, calibration, formulation tests, and physical validation answer different questions. Unit and integration tests can establish that constraints conserve inventory, enforce asset limits, preserve units, and route electrical demand correctly. Feasible MILP residuals establish consistency with the implemented surrogate. Replaying the optimized decisions in independent nonlinear models assesses physical fidelity on the selected instance. A claim about robust real-world operation would require further evidence, including calibrated network data and sensitivity to uncertain forecasts. These stages should remain explicit when interpreting the numerical results.

8 Conclusion

This paper presents a deterministic MILP implementation for centralized short-term water–energy coordination. The model links feasible pump status decisions to electrical bus demand, combines water storage with PV and battery dispatch, and includes approximate hydraulic and electrical network constraints. Consistent interval-end inventories and single-count electricity pricing make the resulting schedules and objective values interpretable. The illustrative experiment finds a small procurement-cost reduction through rescheduling, with unchanged modeled import energy. Nonlinear replay satisfies sampled pressure and voltage limits but reveals a terminal water-reserve shortfall; this discrepancy limits the operational interpretation of the MILP schedules. These conclusions apply to the documented deterministic approximations and case data. Further work should prioritize simulation-informed hydraulic refinements, improved pump-power modeling, and broader network and uncertainty studies before extending those conclusions to operational deployment.

Data and code availability

Network inputs, configuration, optimization code, tests, and experiment-generation scripts are included in the accompanying repository. The manuscript’s numerical section and figures are generated from the experiment outputs so that reported values can be regenerated with the documented commands. The bundled MILPNet reference code is attributed to Thomas and Sela [2]; it is distinct from the project implementation. No confidential measurements or external market datasets are required for the illustrative case. From the repository root, `python -m src.experiments` regenerates the study using `data/inputs/paper_case.yaml`; from the `paper` directory, `latexmk -pdf main.tex` rebuilds the manuscript.

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